

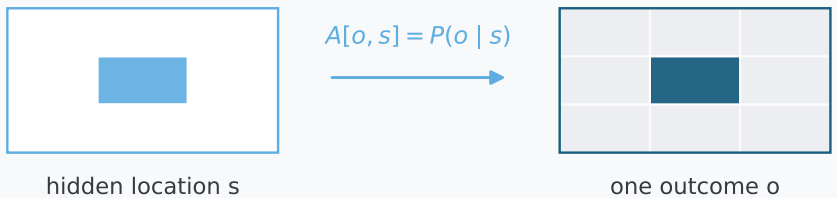
Categorical sentinel generative model and federated belief paths

Friston-style hidden location, private observation, local posterior, and optional control depth

$$q(s) = \text{softmax}(\ln D_0(s) + \ln A[o, \cdot]) \quad | \quad \text{robust_aggregate}(0) = \text{log_linear_pool}$$

A What each sentinel sees

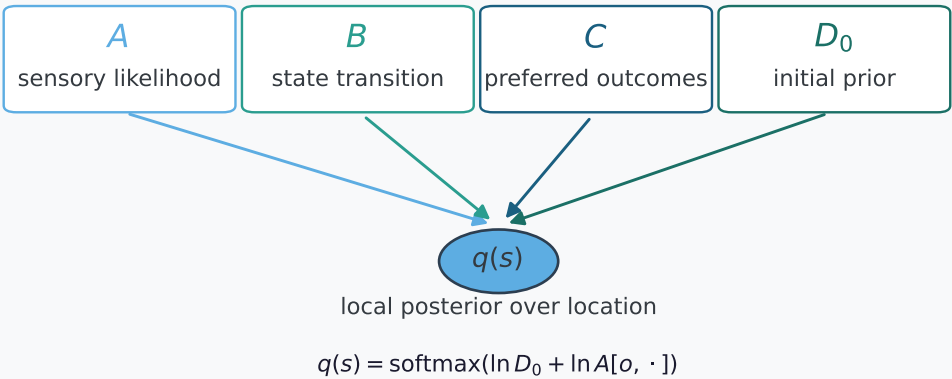
A private categorical report over the shared nine-cell location space



diagonal mass = acuity; residual mass = categorical noise
The raw outcome stays with the agent; only $q(s)$ is broadcast.

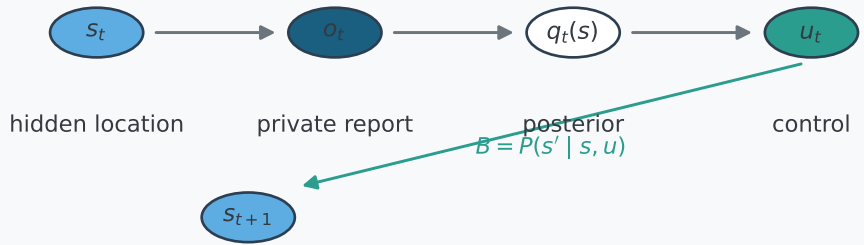
B A/B/C/D_0 model factors

Likelihood, dynamics, preferences, and prior shape local inference and action



C Temporal depth

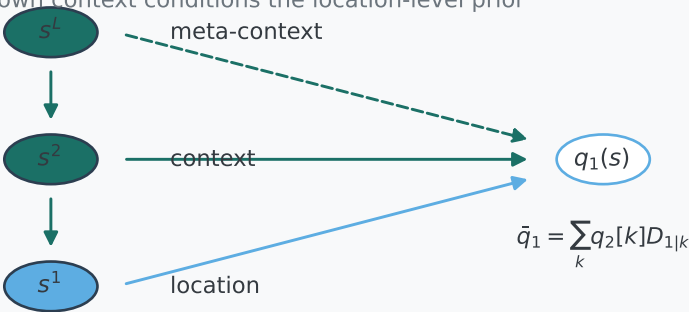
Inference and control across successive times



Flat studies stop after posterior sharing; moving-world studies execute B and EFE-guided u .

D Hierarchical context

Optional top-down context conditions the location-level prior



The hierarchy is an extension, not a hidden assumption in every study.

Schematic formalization: the agent's private sensory outcome becomes a posterior message; the server fuses messages over the shared categorical state.